Synthetic data curation at group granularity

- Effective Synthetic Data Curation Requires Group-Level Signals
- Cathy Jiao and Chenyan Xiong, arXiv v1, September 30, 2026
- Technical reading for AI data infrastructure engineering

> 核心问题：单条样本分数的求和能否代表整组训练价值？本文用受控实验说明二者可能偏离。

Paper v1

The unit of selection changes the objective

- Individual influence estimates a sample's utility in isolation.
- A group changes the model through multiple interacting optimization steps.
- Synthetic paraphrases share origins and may concentrate training signal despite high individual scores.

> 作者研究训练价值的非加性。不要把 group 简化成按来源分桶后的平均质量分。

Paper v1

Proxy oracle and group oracle

- The proxy aggregates individual sample effects around the reference model.
- The group oracle evaluates the validation-loss change after training on the group.
- Matching proxy scores while varying composition isolates interactions that the additive proxy misses.

> Oracle 是昂贵的实验参照，不是可直接用于全量生产的免费标签。参见第 2 节与附录 B.1。

Paper v1

Controlled group construction

- BGE embeddings and k-means partition each corpus into 20 semantic clusters.
- Groups span one to 20 clusters while their average individual proxy scores remain matched.
- Gumbel-Top-k sampling selects groups under matched training budgets.

> 既控制平均单样本效用，又改变语义与主题组成，避免仅用更高质量数据解释结果。第 3.2 节。

Paper v1

Training settings and controls

Setting	Model and data	Control
Continued pretraining	Repro-400M, synthetic Repro corpus	Same initialization and optimization
Post-training	Qwen2.5-1.5B-Instruct, GooseReason	Same step count and evaluation
Selection arms	Group oracle, proxy oracle, random	Same candidate pool and budget

> 这里只覆盖 400M/1.5B 文本实验，不是多模态或前沿规模的直接证据。第 3 节。

Paper v1

Synthetic recycling increases redundancy pressure

- The pool contains 716,800 sequences for every rephrase factor.
- Rephrase factors 1, 10, and 50 reduce unique source documents as repetitions rise.
- FineWeb-Edu, BM25, and GradSim represent individual scoring. GMRel and GREATS include relations.

> r=50 是高合成度压力测试。固定序列总数使来源覆盖度的变化更容易解释。第 4 节。

Paper v1

Reported average accuracy at r = 10

Selection	Average score	Interpretation
Random	29.58	Reference selection
GradSim	28.04	Individual gradient similarity
GMRel	30.40	Relational scoring
GREATS	30.27	Group gradient method

> 表 9 的 8 项任务平均值，200 步 Repro-400M continuation。不同任务表现差异很大，平均分不能替代生成能力分项。

Paper v1

Gradient diversity targets expensive audits

- Within-group gradient diversity predicts the proxy–oracle gap.
- Reported Spearman correlations range from 0.58 to 0.63 across tested settings.
- Audit high-diversity groups with the group oracle and use a cheaper proxy elsewhere.

> 这是预算分配诊断，不是梯度越多样训练价值就一定越高。高 gap 意味着需要更准确估计。第 5 节。

Paper v1

Limits of the evidence

- The oracle depends on the chosen validation objective and training horizon.
- Groups are deliberately constructed. Real production mixtures may have different interactions.
- Gradient extraction adds compute and memory overhead even when subsampling helps.

> 作者结果支持关系信号值得保留，但不能推出任何 diversity 策略都能提升任意训练任务。

Paper v1

Curation data model implications

- Inference: preserve synthetic parent IDs and generator versions alongside each record.
- Inference: materialize group membership and selection runs rather than only a scalar quality column.
- Proposed application: store group diagnostics and downstream evaluation deltas as versioned artifacts.

> 工程建议由机制推导，并非论文交付的产品功能。需要可追踪 group 才能复查选数决策。

Paper v1

A bounded reproduction plan

- Rebuild a small proxy-matched pool and compare random, GradSim, and one group method.
- Fix selected tokens, training steps, model initialization, and held-out tasks across arms.
- Measure generation metrics, oracle rank recovery, gradient memory, and audit cost separately.

> 先验证选择信号是否可迁移，再投资全量梯度计算。本文阅读未执行训练，也未验证专用公开代码仓库。

Paper v1

Adoption decision

- Published claim: group interactions explain selection failures in the tested synthetic settings.
- Inference: per-record scores alone provide an incomplete selection contract.
- Proposed gate: adopt a group diagnostic only after gains survive matched token and compute budgets.

> 决策标准是训练效用与成本共同改善，不能只看离线分数相关性。来源入口及逐页讲解见 sources.md。

Paper v1